Lazizbek. MATH 548. HW # 5.

Solve the following equations using the fixed point iteration method. In each case the starting values are 2, 0.8, 0.2.

Pr. 1.

(1) $x = x^2 \quad \Rightarrow \quad x = g(x) \quad \Rightarrow$

(2) $g(x) = x^2$.

We know that for x_n

(3) $$x_{n+1} = x_n^2$$

fixed point iteration to be convergent

(4) $$|g'(x)| < 1$$

If we solve this (4), we find the convergent interval of (3):

(5) $|g'(x)| = |2x| < 1 \quad \Rightarrow \quad -1 < 2x < 1 \quad \Leftrightarrow$

(6) $x \in (-\frac{1}{2}, \frac{1}{2})$ = convergent interval for (3).

So if we start $x_0 \in (-0.5, 0.5)$ then (3) converges otherwise diverges, also $|g'(x_0)|$ can tell us how fast/slow (3) converges or diverges:

Let $x_0 = 2 \notin (-0.5, 0.5)$, so we can see

(7) $x_{n+1} = x_n^2$, $x_0 = 2$ diverges

(8) $|g'(x_0)| = |g'(2)| = |2x_0| = |2 \cdot 2| = 4 > 1 \Rightarrow$ (3) diverges.

(9) $\begin{cases} x_0 = 0.8 \notin (-0.5, 0.5), \text{ so} \\ x_{n+1} = x_n^2, \quad x_0 = 0.8 \text{ diverges as} \end{cases}$

(10) $|g'(x_0)| = |2x_0| = |2 \cdot 0.8| = 1.6 > 1 \Rightarrow$ (3) diverges.

yet,

(11) $\begin{cases} x_0 = 0.2 \in (-0.5, 0.5) \\ x_{n+1} = x_n^2, \quad x_0 = 0.2 \text{ converges as} \end{cases}$

(12) $|g'(x_0)| = |2x_0| = |2 \cdot 0.2| = 0.4 < 1 \Rightarrow$ (3) converges.

Rate of convergence/divergence is determined by

$4 > 1.6 > 1 \iff$

(13) $\begin{cases} |g''(2)| = 4 > 1 \Rightarrow (7) \text{ diverges faster than } (9), \\ |g'(0.8)| = 1.6 > 1 \Rightarrow (9) \text{ diverges slower than } (7), \\ |g'(0.2)| = 0.4 < 1 \Rightarrow (11) \text{ converges, nothing to compare.} \end{cases}$

You can see the number of steps to converge or diverge within the given tolerance in the program, attached at the end.

For the rest of the problems, let me not to be so detailed where I'll be calculating only $|g'(x_0)|$ to identify the rate of convergence/divergence to compare with the program:

Pr2.

$$x = \sqrt{x}$$

(14) $x_{n+1} = \sqrt{x_n} \Rightarrow$

(15) $g(x) = \sqrt{x} \Rightarrow$

(16) $g'(x) = \dfrac{1}{2\sqrt{x}} \Rightarrow$

(17) $|g'(x)| = \left|\dfrac{1}{2\sqrt{x}}\right| < 1 \Leftrightarrow -1 < \dfrac{1}{2\sqrt{x}} < 1 \Leftrightarrow \overbrace{-2 < \dfrac{1}{\sqrt{x}} < 2}^{x \neq 0,\, x \in R} \Rightarrow$

$\Leftrightarrow \dfrac{1}{\sqrt{x}} < 2 \Leftrightarrow \sqrt{x} > \dfrac{1}{2} \overset{(18)}{\Leftrightarrow} \boxed{x > \dfrac{1}{4}}$ convergence interval for (14).

(19) $\begin{cases} x_0 = 2 \in \left(\frac{1}{4}, \infty\right) \Rightarrow (14) \text{ converges "faster"} \\ |g'(x_0)| = \left|\dfrac{1}{2\sqrt{x_0}}\right| = \left|\dfrac{1}{2\sqrt{2}}\right| = 0.35 < 1 \end{cases}$

(20) $\begin{cases} x_0 = 0.8 \in (0.25, \infty) \Rightarrow (14) \text{ converges "slower"} \\ |g'(x_0)| = \left|\dfrac{1}{2\sqrt{0.8}}\right| = 0.56 < 1 \end{cases}$

(21) $\begin{cases} x_0 = 0.2 \notin (0.25, \infty) \Rightarrow (14) \text{ diverges.} \\ |g'(x_0)| = \left|\dfrac{1}{2\sqrt{0.2}}\right| = 1.11 > 1 \end{cases}$

Rate of convergence / divergence is as followse

$0.35 < 0.56 < 1 \Rightarrow$ (19) converges faster than (20).

Pr 3.

(22) $x = x + 2(x - x^2) \Rightarrow$

(23) $x_{n+1} = x_n + 2(x_n - x_n^2) \Rightarrow$

(24) $g(x) = x + 2(x - x^2) = x + 2x - 2x^2 = 3x - 2x^2$

(25) $g'(x) = 3 - 4x$

(26) $|g'(x)| = |3 - 4x| < 1 \iff -1 < 3 - 4x < 1 \Rightarrow$

(27) $\Rightarrow \boxed{\frac{1}{2} < x < 1}$ convergence interval for (23).

(28) $\begin{cases} x_0 = 2 \notin \left(\frac{1}{2}, 1\right) \Rightarrow (23) \text{ diverges "faster"} \\ |g'(x_0)| = |3 - 4x_0| = |3 - 8| = 5 > 1 \end{cases}$

(29) $\begin{cases} x_0 = 0.8 \in \left(\frac{1}{2}, 1\right) \Rightarrow (23) \text{ converges} \\ |g'(x_0)| = |3 - 4x_0| = |3 - 4 \cdot 0.8| = 0.2 < 1 \end{cases}$

(30) $\begin{cases} x_0 = 0.2 \notin \left(\frac{1}{2}, 1\right) \Rightarrow (23) \text{ diverges "slower"} \\ |g'(x_0)| = |3 - 4 \cdot 0.2| = 2.2 > 1 \end{cases}$

Rate of convergence/divergence

$5 > 2.2 > 1 \Rightarrow$ (28) diverges faster than (30).

Pr. 4.

(31) $x = x - \dfrac{x - x^2}{1 - 2x} \Rightarrow$

(32) $x_{n+1} = x_n - \dfrac{x_n - x_n^2}{1-2x_n} \Rightarrow$

(33) $g(x) = x - \dfrac{x - x^2}{1-2x}$

(34) $|g'(x)| = \left| 1 - \dfrac{(1-2x)(1-2x) - (x-x^2)\cdot(-2)}{(1-2x)^2} \right| < 1 \Leftrightarrow$

(35) $\left| \dfrac{(1-2x)^2 - (1-2x)^2 - 2(x-x^2)}{(1-2x)^2} \right| = \left| \dfrac{-2(x-x^2)}{(1-2x)^2} \right| < 1$

As (35) is difficult to solve directly, let me calculate & check $|g'(x_0)| < 1$ separately:

(36) $\begin{cases} x_0 = 2 \\ |g'(x_0)| = \left| \dfrac{-2(x_0 - x_0^2)}{(1-2x_0)^2} \right| = \left| \dfrac{2(2-4)}{(1-4)^2} \right| = \left| \dfrac{4}{9} \right| = \dfrac{4}{9} < 1 \end{cases}$

$\Rightarrow$ (32) converges at $x_0 = 2$ faster than $x_0 = 0.8$, $x_0 = 0.2$:

(37) $\begin{cases} x_0 = 0.8 \\ |g'(x_0)| = \left| \dfrac{-2(x_0 - x_0^2)}{(1-2x_0)^2} \right| = \left| \dfrac{-2(0.8 - 0.8^2)}{(1-2\cdot 0.8)^2} \right| = \left| \dfrac{2\cdot 0.16}{0.36} \right| = \dfrac{8}{9} < 1 \end{cases}$

$\Rightarrow$ (32) converges

(38) $\begin{cases} x_0 = 0.2 \\ |g'(x_0)| = |g'(0.2)| = \left| \dfrac{-2(0.2 - 0.2^2)}{(1-2\cdot 0.2)^2} \right| = \left| \dfrac{2\cdot 0.16}{0.36} \right| = \dfrac{8}{9} < 1 \end{cases}$

(32) converges

$\dfrac{4}{9} < \dfrac{8}{9} = \dfrac{8}{9} < 1 \Rightarrow x_0 = 2$ has a faster convergence.